

RE²_nAF: EXPLORING SCENARIOS FOR OFF GRID ELECTRICITY PRODUCTION OPTIONS IN AFRICA

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ABSTRACT: The international community is committed to supporting Africa to address its energy challenges but the definition of continental priorities and ways of implementation is under national, sub-regional and regional African policy domains. The scope of this work is to present an analytical easy-to-use web-based tool, RE²_nAF, which enables geographically based analysis for off-grid electricity production options in Africa. Decision makers, academia, investors and other stakeholders can benefit from its use in deciding which energy technology options would better deliver electrification services to rural areas in Africa, after exploring the impact on least cost option caused by simulating different scenarios.

Keywords: energy options, modelling, off grid systems, rural electrification, Africa, web mapping tool.

1 INTRODUCTION

1.1 Electricity access in Africa

Africa, with 5% of global primary energy use and 15% of the world population, represents only a third of the per capita energy consumption of the global average, explained, among others, by low access rates. According to the statistics, 42.6% of African population has access to electricity, an average figure that hides the gap between North Africa, with 99.4% electrification rate, and Sub-Saharan Africa, with just 31.8% [1]. As part of the same reality, many other studies and reports reveal the inadequacy in electricity provision and the high frequency of power outages (in Sub-Saharan Africa, 30 out of 48 countries experience daily power outages [2]).

In 2011, 600 million Africans lacked access to electricity; Sub-Saharan Africa accounted for 99.8%, being the vast majority of non-electrified population living in rural areas.

Despite the success of some electrification programmes (mostly in North Africa), the commitment of the international community and the prevailing role of energy in the fight for economic and social development, a much greater effort must be done for the electrification rates to catch up with the population growth. Achieving universal access by 2030 in Africa (goal set by UN on 2012 as the International Year of Sustainable Energy for All and further pursue in post-agenda 2015 through the declaration of 2014-2024 as the Decade of Sustainable Energy for All) implies providing electricity to 1.5bn people [3] and above all, relies on the implementation of an adequate roadmap and a set of concrete targets.

1.2 Uncertainty and rural electrification

Rural electrification planning is a complex matter in which the definition of a time horizon, the objective of increasing access to energy in a given territory (minimizing costs) and considering constraints that could be political, technical, financial and/or strategic, are all factors with great influence.

Spatial particularities (concerning the resources available, demand, technology, local expertise, etc.) play an important role in defining the most appropriate electrification technology. Decisions must be taken upon the consideration of such particularities so rural electrification initiatives could be implemented and last longer in situations where economic resources are limited.

Due to the number of different variables (also of different nature) the use of models and tools leads to improved quality in energy planning decision making by facilitating the simulation of scenarios. In most cases, the values of those variables are affected by unpredictable events whose consequences may change the competitiveness of possible alternatives completely. Some level of uncertainty is inherent to rural electrification planning, and therefore, simulation is needed to assess the magnitude of changes in the initial assumptions. Accurate description of the renewable resources potential (spatially distributed, based on long time series) is not always available. Technology is continuously evolving, improving efficiencies and increasing lifetime. Changes in the energy market cause variations in the cost of technology, components, labor or fuel. Other conditions might affect the socio-economic scenario, affecting consumption, load density and discount rate among others.

1.3 Purpose

Several studies have concluded that renewable energy is often the cheapest solution for off-grid solutions in many developing countries [4], [5]. But is not so common to find in the bibliography analysis that present, at continental scale, the spatially distributed competitiveness for rural electrification technologies [6]. The results showed, under established assumptions, the huge potentiality of PV in Africa remote areas and the necessity of assessing the different electrification technologies simultaneously. The methodology described in [6] has been used to create the geoeconomic layers contained in the web mapping tool described in detail in this paper (for more information on the methodology, please refer to [6], [7]): RE²_nAF (accessible via <http://re.jrc.ec.europa.eu/re2naf.html>; contact email re2naf@jrc.ec.europa.eu)

Informed decision making (in a general sense: ranging from National Governments to local investors) depends on the availability of reliable input data, appropriate techniques to simulate such data (models, software, means of validation, etc.) and a trained user community. RE²_nAF's interface (**figure 1**) has been designed to meet the needs of all different types of users. For non-experts, the interface provides easy navigation through the default datasets, enables to manipulate its visualization according to their needs, and to perform basic queries (see Section 2). For more advanced users,

declares the difference in fuel prices due to transportations costs, being inland prices higher than in coastal areas. According to the same source, “(...) the cost of road delivery from the depot towns to the respective destinations is added which may vary considerably from about 5 cents per kilometer to 16 cents per kilometer as it is a function of distance, quantity and type of contractor”. The government subsidizes the transportation costs to the oil companies. According to [11], also across Lesotho (almost 28 times smaller than Namibia) the fuel prices differ (four pricing zones), probably due to transportation costs.

Table II summarizes the spatial variability of diesel based costs per country for three different diesel prices (those reported for 2008, 2010 and 2012 [9]), including minimum, mean and maximum value per country depending on the diesel price considered in the scenario.

Table II: Electricity based in diesel minigrid: minimum, average and maximum costs in Southern African countries (c€/kWh)

Country	2008	2010	2012
	m; av; mx	m; av; mx	m; av; mx
Botswana	22; 51; 155	21; 46; 138	29; 63; 190
Lesotho	22; 38; 79	23; 41; 84	30; 53; 109
Namibia	15; 38; 82	15; 44; 95	16; 55; 121
South Africa	22; 30; 86	20; 33; 82	30; 44; 109
Swaziland	22; 27; 59	20; 30; 52	30; 39; 67

* ‘m’ stands for minimum; ‘av’ for mean and ‘mx’ for maximum. All values have been rounded up to the nearest unit.

3.2 PV based electricity costs

In order to better reflect local differences on PV module market prices, the user can select among 8 different values ranging from 0.75 to 2.5 €/W_p. Once a value is chosen, the “Add Layer” button is enabled allowing the user to visualize the spatial distribution of the cost of generating 1 kWh by means of PV minigrid in the selected country. It is also possible to get the cost for a particular location. PV costs are always represented in a yellow-red color scheme. For these calculations, the module price and the maintenance costs (2.5% per year of the combined cost of modules plus BOS) vary. Just the BOS is constant and fixed to 1€/kW_p. Consult the following reference for further information [12].

Table III summarizes the spatial variability of PV based costs per country for three selected module prices (0.75€/W_p, 1.5€/W_p, 2.5€/W_p), including minimum, mean and maximum value per country depending on the scenario.

Table III: Electricity based in PV minigrid: minimum, average and maximum costs in Southern African countries depending on PV module price (c€/kWh)

Country	0.75€/W _p	1.5€/W _p	2.5€/W _p
	m; av; mx	m; av; mx	m; av; mx
Botswana	22; 25; 30	26; 32; 40	33; 42; 52
Lesotho	29; 31; 38	36; 40; 49	45; 51; 65
Namibia	22; 30; 52	24; 28; 43	31; 36; 57
South Africa	22; 30; 52	28; 38; 68	36; 50; 90
Swaziland	29; 32; 37	36; 41; 47	45; 52; 61

* ‘m’ stands for minimum; ‘av’ for mean and ‘mx’ for maximum. All values have been rounded up to the nearest unit.

3.3 Comparison

When least cost option technology is desired, comparison among the alternatives is needed (currently RE²nAF includes diesel-PV comparison). Because the real potential of each option will be determined not only by minimizing the cost per kWh but also by the difference between its own costs and those of the other technologies which are in direct competition with it.

Using RE²nAF, and after diesel and PV parameters have been set, the user can explore the output resulting from the comparison. The value included in this layer is the computed difference between the PV and the diesel electricity cost (in cts€/kWh) for a given location. The symbology has been defined to help the user interpreting the results: yellow-red color scheme is applied to areas where PV is the least cost option while blue is applied to the areas where diesel is the cheapest.

The following figures show the spatial competitiveness of the two technologies for given scenarios: **figure 3** (scenario a) corresponds to a calculation based on 2012 diesel price and PV module price of 2.5 €/W_p while **figure 4** (scenario b) corresponds to a calculation based on 2012 diesel price as well and PV module price of 1.5 €/W_p. Both maps clearly reflect the influences that changes in input parameters (in these cases, PV module price; but every single variation in input values would change the picture to some degree) have in least cost electricity option, globally and also (and more importantly) locally.

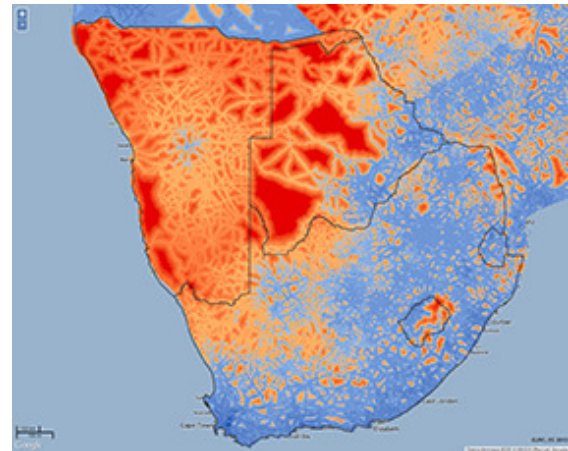


Figure 3: Scenario a) Comparison diesel vs. PV minigrid (PV module price 2.5 €/W_p; 2012 diesel price)

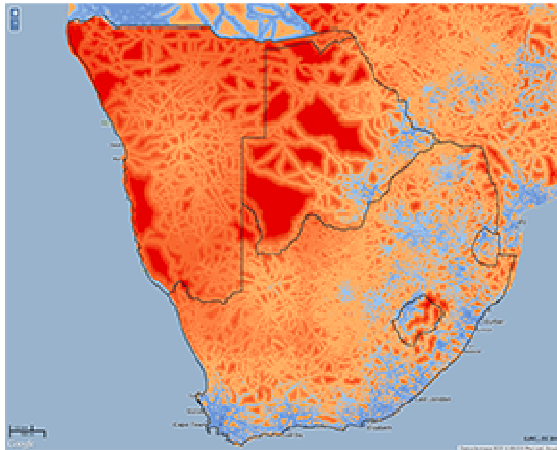


Figure 4: Scenario b) Comparison diesel vs. PV minigrid (PV module price 1.5 €/W_p; 2012 diesel price)

A third scenario (c) has been evaluated (diesel prices are maintained, while PV module price is set to 0.75€/W_p). The values that illustrate the changes in minimum, average and maximum cost are included in **table III**. On the other hand, **figure 5** shows the percentage of population allocated to least cost electricity option per scenario. In all five countries, a reduction of 1€/W_p (difference between scenario a and scenario b) increases in more than 15% the population living in areas where PV is the cheaper. Namibia is the only case in which this reduction makes PV the least cost option for the whole population. PV module price set to 0.75€/W_p (scenario c) makes PV the cheapest option for at least 80% of the population.

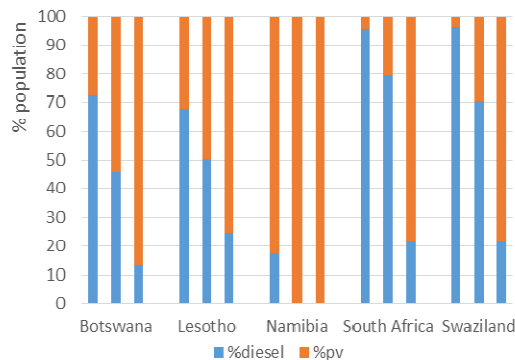


Figure 5: Population allocated to least cost electricity option, comparison of scenarios per country

* First column: scenario a; second column: scenario b; third column: scenario c.

4 CONCLUSIONS

Renewable energy technologies such as micro and mini hydro, PV minigrids, hybrid wind-PV solutions, etc. are proven technologies that have provided access to energy in vast off-grid areas worldwide. Renewable energy technologies are not just a critical element of the solution to climate change, but a key instrument to develop local economy, increase energy independence, and achieve sustainable development and energy access for all. A stable policy framework backed by strong political will, local industry and educated labor force are

necessary to catalyze action towards achieving energy access for all. National policies should contribute towards creating a cost-effective and affordable energy market.

Decision support tools can provide valuable feedback and information to policy makers aiming to design and implement effective energy access programs serving growing population in Africa. This tool is in line with the current international political momentum around access to energy and it will strengthen the capacities of African rural electrification institutions, furthering the goal of expansion of electricity and renewable energies. A change in paradigm from considering just electricity network extension towards including as potential option small scale renewable solutions is imperative.

The tool described in this paper, RE²nAF, is a Web Mapping Application that will allow the user, GIS skilled and non-skilled, with means to consult spatially represented data with no requirement of specialized software or downloads.

Although many improvements are planned for RE²nAF in the near future (such as the incorporation of other renewable sources, costs of grid-connected PV systems, wider range of values for users to simulate further scenarios, etc.), it currently enables the interaction with data regarding renewable energy potentials, spatial distribution of energy demand, existing infrastructure and other relevant geographical information. It also offers the possibility to assess renewable energy and electrification policies impact by simulating optimistic and/or conservative scenarios.

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6 ACKNOWLEDGEMENTS

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RE²_nAF:

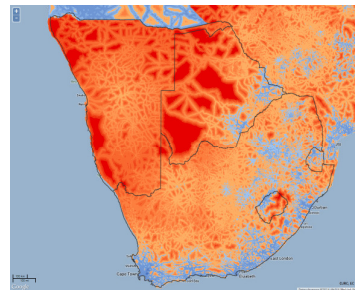
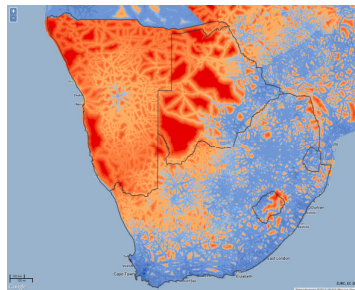
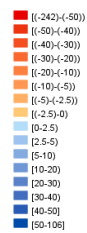
exploring scenarios for off grid electricity production options in Africa

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Why? SCENARIOS

Difference PV–
diesel costs
[cts€/kWh]



Why?

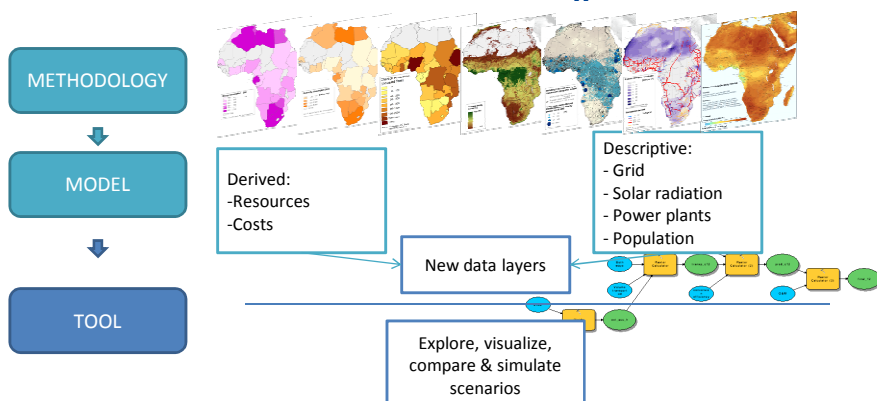
What is the global irradiation in Windhoek?

How much does it cost 1kWh produced by PV in Goundam, Mali?

How far is Garissa (Kenya) to the nearest transmission line?

After setting a diesel and PV module price, which off-grid technology is cheaper in Sokode, Togo?

From methodology to RE²_nAF



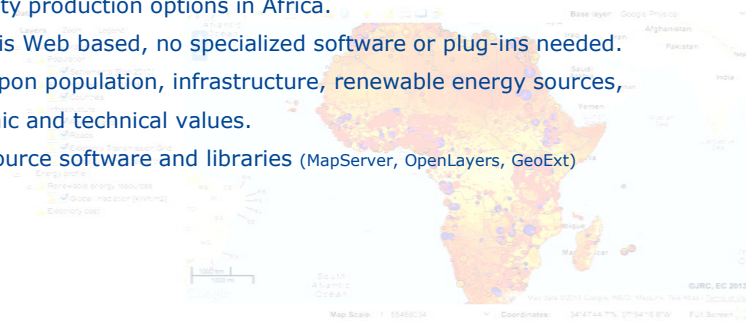
Maps: Monforti-Ferrario (ed). (2011) Renewable Energies in Africa

Methodology: Szabó, S., Bódis, K., Huld, T., & Moner-Girona, M. (2011). Environmental Research Letters, 6(3), 34002.

What is RE²_nAF?

RE²_nAF renewable energies
rural electrification
africa

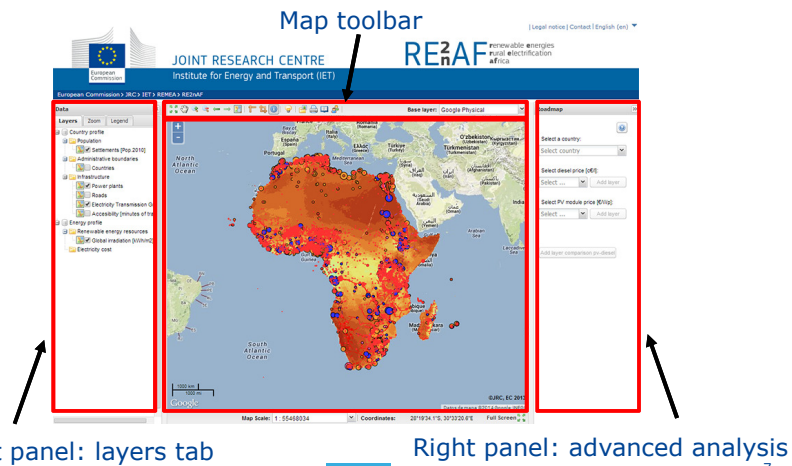
- Enables geographically based exploratory analysis for off-grid electricity production options in Africa.
- RE²_nAF is Web based, no specialized software or plug-ins needed.
- Builds upon population, infrastructure, renewable energy sources, economic and technical values.
- Open source software and libraries (MapServer, OpenLayers, GeoExt)



RE²_nAF processes data

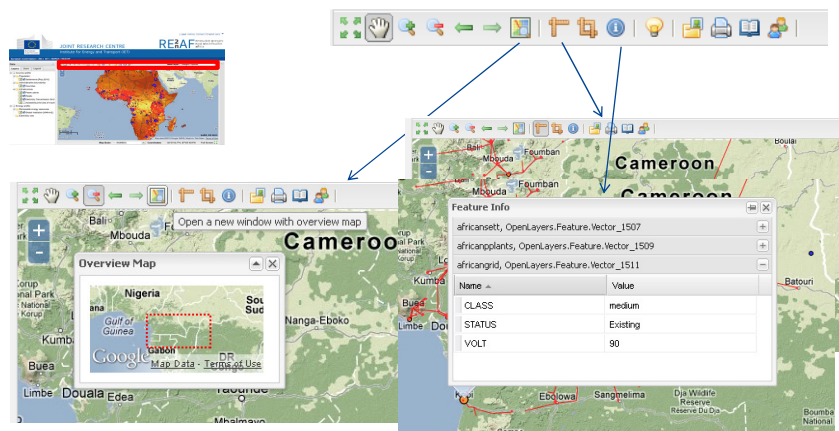
- Settlements: > 6,700
(population data 2010, less than half of estimated total population in Africa of more than 1000M)
- Electricity grid: > 230,000 km in 46 countries (~40% existing)
- Power plants: >900 in 54 countries (~200 GW)
- Global irradiation: ~900 to >2600kWh/m²
- Diesel: 2008, 2010 and 2012 estimated costs 3 - > 250 cts€/ kWh
- PV: 8 layers (using module prices from 0.75 to 2.5 €/W_p) with estimated costs of 17-112 cts€/kWh

Basic interaction



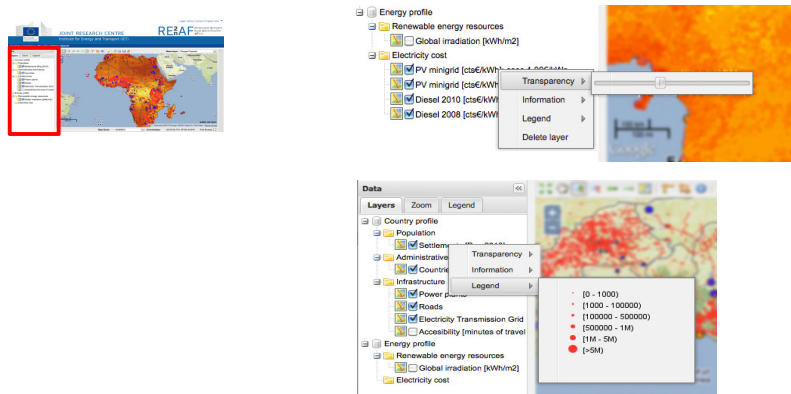
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Map toolbar

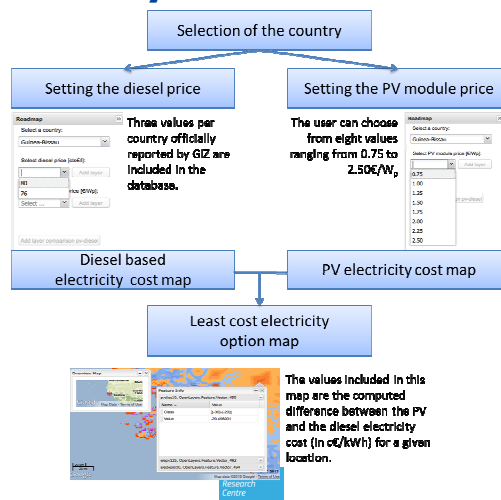


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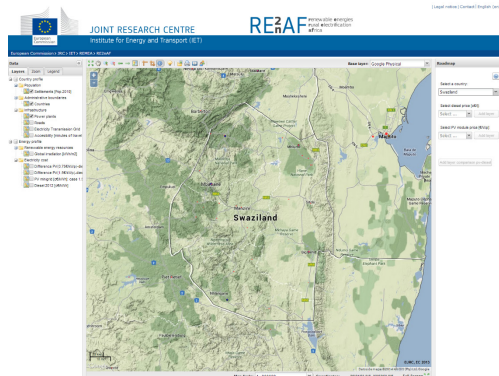
Right panel: layers tab



Advanced analysis

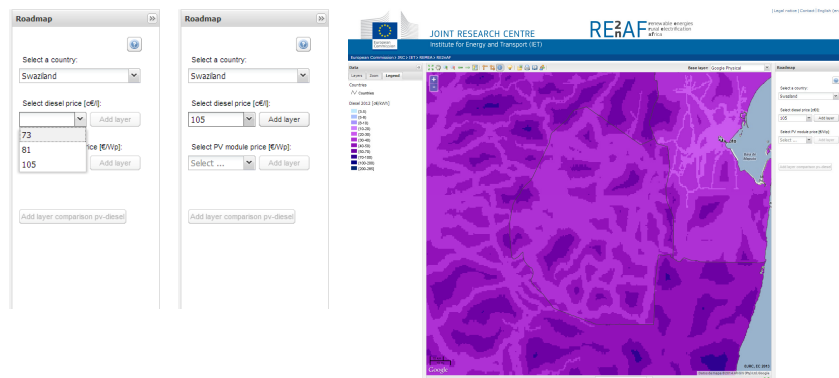


Advanced analysis

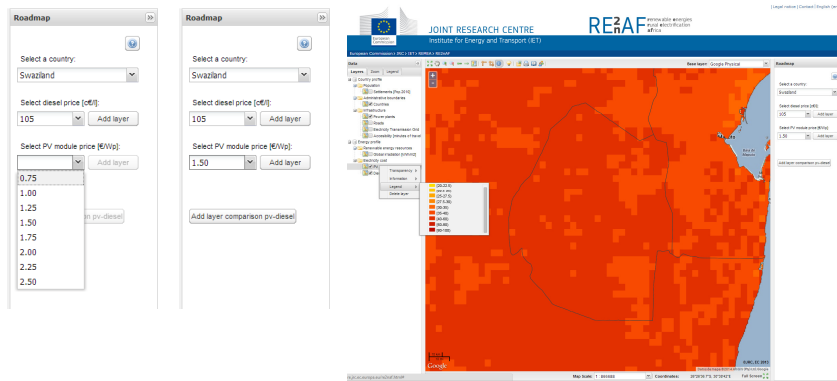


Swaziland:

Diesel



Photovoltaic

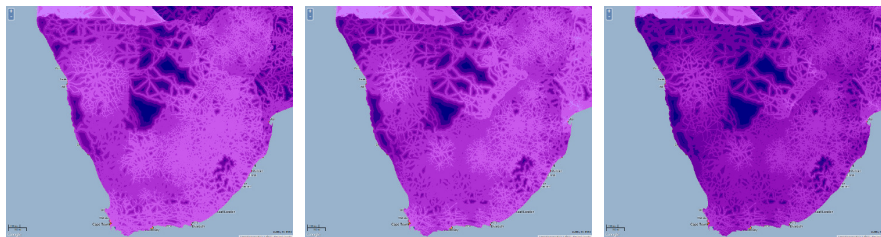
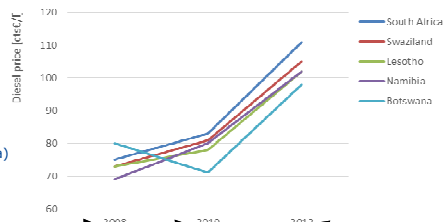


Comparison

Southern African region (UN def.),

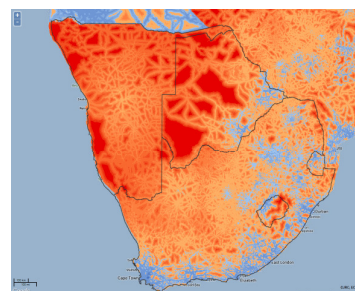
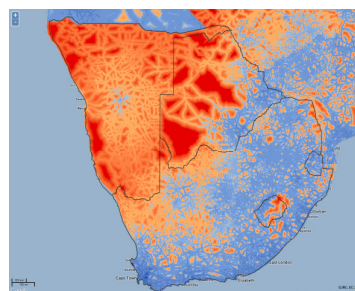
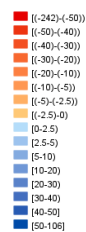
11M people without electricity

Rural electrification rate 9% (Lesotho) – 67% (South Africa)



Comparison

Difference PV–
diesel costs
[cts€/kWh]



Southern African region (UN def.), comparison off-grid PV-diesel based

Left: 2012 diesel price
PV price 2.5€/kWp

Right: 2012 diesel price
PV price 1.5€/kWp

Final remarks:

- Can be used by everybody: decision makers , entrepreneurs, academia. User friendly.
- Importance of reliable, accurate & updated spatially distributed information.
- Enables calculation of “*real*” cost of renewables for local conditions.



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RE²NAF

European Commission > IET > IET > REMEA > RE2NAF

Layers

- Country profile
- Population
- Settlements (Pop 2010)
- Infrastructure
 - Power plants
 - Roads
 - Electricity Transmission Grid
 - Accessibility (Indices of travel)
- Energy profile
 - Renewable energy resources
 - Global irradiation (potential)
 - Electricity cost

Base layer: Google Physical

Map Scale: 1:55468004 Coordinates: 34°47'44" N, 07°54'18" W Full Screen

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